Multi-LLM Chat Client – OpenAI, DeepSeek, Google Gemini, Hugging Face & GroqCloud

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Kompressor — Intelligent Context Compression

Kompressor is Headroom's context compression engine. It automatically detects large tool outputs, file reads, and search results in your conversation history and compresses them before they reach the LLM — saving tokens and money without losing the information the model needs.

The implementation lives in `headroom/transforms/kompressor.py`.

Core Concept

Every message in the conversation is scanned. If a message's **estimated token count** exceeds the configured threshold, Kompressor compresses it. The compressed version replaces the original in the message list sent to the LLM. The original is never modified — Kompressor operates on a copy.

Activation Thresholds

Provider	Threshold
Anthropic (Claude)	10,000 tokens
OpenAI (GPT-4, etc.)	10,000 tokens
Google (Gemini)	10,000 tokens
Default	10,000 tokens

Override with `--kompressor-threshold N` or `HEADROOM_KOMPRESSOR_THRESHOLD=N`. Set to `0` to disable Kompressor entirely.

Compression Process (Step by Step)

1. Scan each message's estimated token count.
2. If below threshold → skip, message passes through unchanged.
3. If above threshold → check if the message is a tool result, file read, or search output.
4. Detect the format (JSON, code, log, plain text) and choose the appropriate compressor.
5. Run the compressor, targeting a reduction ratio (default: keep ~30% of tokens).
6. Replace the message content with the compressed version.
7. Record the savings in the transform result for metrics tracking.

Smart Summary Discard

When Kompressor compresses a message, it doesn't just truncate. It **summarizes** the discarded portion — keeping the first few lines, the last few lines, and a structural summary of what was in the middle. This preserves the model's ability to understand the overall shape of the data.

Provider Restriction (Paid Only)

Kompressor is restricted to paid providers (Anthropic, OpenAI, Google) by default. Free-tier providers (Groq, Together, Ollama) are excluded because the compression overhead can actually increase latency on fast, cheap models.

Provider	Kompressor Enabled?
Anthropic	Yes — Claude models
OpenAI	Yes — GPT-4, GPT-3.5
Google	Yes — Gemini models
Groq, Together, Ollama, etc.	No — free tier

Override with `--kompressor-all-providers` or `HEADROOM_KOMPRESSOR_ALL_PROVIDERS=true`.

Result Files

Each compression result is stored in memory with:

- Original token count
- Compressed token count
- Compression ratio
- Format detected
- Compressor used

Quota & Limit Banners

Free tier — "You've used 80% of your daily limit. Upgrade to Pro for unlimited access." Dismissible, reappears next session.

Pro tier — "You've reached your monthly limit. Overage billing is active." Non-dismissible, persistent.

Context Window Exceeded Handling

When the Anthropic API returns a context_length_exceeded error, the proxy automatically:

1. **Parses the error** — Extracts the actual token count and the limit from the error message.
2. **Trims messages** — Removes the oldest non-system messages until the request fits.
3. **Preserves system messages** — Never trims system prompts or the most recent user message.
4. **Retries once** — Sends the trimmed request. If it fails again, returns the error to the client.
5. **Records the event** — Logs how many messages were dropped and how many tokens were saved.

The trimming strategy is conservative — it drops the minimum number of messages needed.

Streaming responses are handled identically — the error is caught before any SSE events are sent.

API Proxy Documentation Headers

Every response from the proxy includes custom headers that describe what happened to your request:

- **X-Headroom-Tokens-Before** — Token count before compression
- **X-Headroom-Tokens-After** — Token count after compression
- **X-Headroom-Transforms** — Comma-separated list of transforms applied

These headers are visible in browser DevTools, curl -v output, and any HTTP client that inspects response headers.

Download Mode

The dashboard supports downloading individual sessions as JSON files. Click the download button on any session in the Sessions tab.

- **Single session** — Downloads one session as a pretty-printed JSON file.
- **Filtered export** — Downloads all sessions matching the current search/filter as a JSON array.
- **Format** — Uses the same schema as `GET /sessions/{id}`.

Responsive Design

The dashboard is fully responsive and works on desktop, tablet, and mobile.

- **Desktop (≥ 1024px)** — Full layout with all columns visible. Charts render at full size.
- **Tablet (768px – 1023px)** — Compact layout. Secondary columns hidden. Charts scale down.
- **Mobile (< 768px)** — Single-column layout. Tables become scrollable. Navigation collapses into a hamburger menu.

DeepSeek V4 Migration

Background

New Models

Architecture Improvements (V4 vs. V3)

Deprecation Timeline

Changes Made in This Project (11 May 2026)

API Compatibility

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19 July 2026 Maintenance & Feature Update

1. Critical Bug: `deepseek-v4-pro` Was Never Reachable

2. Diagnostic Logging: Model Name in Server Log

3. Dead Model Cleanup

4. OpenAI Lineup Updated to GPT-5.5 / GPT-5.6

5. Transparent API Error Messages

6. `max_tokens` → `max_completion_tokens` for OpenAI

7. Vision Pipeline Completed

8. `deploy.sh`: MD5 Checksum Verification

Files Changed

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28 August 2026 Fix: Explicit DeepSeek Thinking Mode Control

Files Changed

Security Hardening (26 July – 15 September 2026)

Stage 1 — Stored XSS and Path Traversal

Stage 2 — CORS Wildcard and Request Size Limits

Stage 3 — Log Exposure and Host Header Injection

Stage 4 — Deliberately Not Implemented: Endpoint Authentication

Stage 5 — Documentation Gap and Deferred Modularization

Residual XSS Follow-Up (14 September 2026)

HTTPS Migration: From Self-Signed to Let's Encrypt

The Fix

A Naming Trap, Caught Before Deployment

Result

Side Effects Found Along the Way

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17 September 2026: Model Display Labels & Cleanup

Removed: `deepseek-v4-flash-vision-exp`

Added: Readable Model Labels with Release Dates

A Bug in the First Pass

The Helper Script `repo2text.sh`

Security Architecture in Detail

1. API Keys — Never Exposed to the Client

2. Magic Byte Inspection

3. Secure Session Storage

4. Log Without Sensitive Data

5. No Direct Client-API Communication

6. Input Validation

7. Transport Security

Deployment & Usage

Prerequisites

Installation

Environment Setup

```
env_name = "ai-model-deployment"
```

Dependency Installation

```
pip install -r requirements.txt
```

Environment Variables

```
# Create a .env file with your API keys and configuration settings
```

Configuration

Model Configuration Create a new model configuration file in the `configs/models/` directory:

```
# Example configuration for a new model
model_name: "my-custom-model"
model_type: "text-generation"
model_path: "/path/to/model/weights"
tokenizer_path: "/path/to/tokenizer"
max_length: 2048
temperature: 0.7
top_p: 0.9
top_k: 50
repetition_penalty: 1.1
batch_size: 8
device: "cuda:0"
precision: "fp16"
quantization:
  enabled: true
  bits: 8
  method: "dynamic"
optimization:
  use_flash_attention: true
  use_kv_cache: true
  compile_model: false
```

Save this configuration as `configs/models/my-custom-model.yaml`.

Environment Variables Set the following environment variables in your `.env` file or system environment:

```
# Model paths
MODEL_CACHE_DIR=/path/to/model/cache
DEFAULT_MODEL=my-custom-model

# API keys (if needed)
HUGGINGFACE_API_KEY=your_huggingface_api_key
OPENAI_API_KEY=your_openai_api_key

# Server configuration
HOST=0.0.0.0
PORT=8000
WORKERS=4
LOG_LEVEL=info

# Performance settings
MAX_CONCURRENT_REQUESTS=100
REQUEST_TIMEOUT=60
```

Deploy Scripts

Script	Purpose
`./scripts/deploy.sh`	Deploys the model server to the specified environment (e.g., `./scripts/deploy.sh production`)
`./scripts/rollback.sh`	Rolls back to the previous deployment if issues are detected
`./scripts/scale.sh`	Scales the number of instances (e.g., `./scripts/scale.sh 5`)
`./scripts/monitor.sh`	Opens the monitoring dashboard

Important: Make sure to **Never Log Secrets** in your code. Use environment variables or secure vaults for sensitive information.

```
# Example of secure logging
logger.info(f"Processing request for user: {user_id}")  # Good
logger.info(f"Using API key: {api_key}")  # Bad - never log secrets!
```

The weights are downloaded from the Hub (~500 GB) and cached locally. Plan accordingly—ensure you have sufficient disk space before starting the download.

Project Structure

```
├── .github/
│   ├── workflows/
│   │   ├── ci.yml
│   │   └── deploy.yml
│   └── ISSUE_TEMPLATE.md
├── configs/
│   ├── models/
│   │   ├── model-a.yaml
│   │   └── model-b.yaml
│   └── server.yaml
├── data/
│   ├── .gitkeep
│   └── sample_input.json
├── docker/
│   ├── Dockerfile
│   ├── Dockerfile.dev
│   └── docker-compose.yml
├── docs/
│   ├── api.md
│   ├── architecture.md
│   └── deployment.md
├── logs/
│   └── .gitkeep
├── models/
│   └── .gitkeep
├── notebooks/
│   ├── 01-exploration.ipynb
│   ├── 02-fine-tuning.ipynb
│   └── 03-evaluation.ipynb
├── scripts/
│   ├── deploy.sh
│   ├── download_model.py
│   ├── rollback.sh
│   └── scale.sh
├── src/
│   ├── __init__.py
│   ├── api/
│   │   ├── __init__.py
│   │   ├── routes.py
│   │   └── schemas.py
│   ├── config.py
│   ├── inference/
│   │   ├── __init__.py
│   │   ├── engine.py
│   │   └── preprocessing.py
│   ├── main.py
│   └── utils/
│       ├── __init__.py
│       ├── logging.py
│       └── metrics.py
├── tests/
│   ├── __init__.py
│   ├── conftest.py
│   ├── test_api.py
│   └── test_inference.py
├── .dockerignore
├── .env.example
├── .gitignore
├── LICENSE
├── Makefile
├── README.md
└── requirements.txt
```

Model Configuration

The server reads its configuration from `configs/server.yaml` and the per-model files in `configs/models/`. Environment variables override values in the YAML files, which is useful for secrets and deployment-specific settings. See `src/config.py` for the full schema and default values.

Adding a new model configuration. Drop a new YAML file in `configs/models/` and restart the server. The model will be available by the filename stem (e.g., `model-c.yaml` → `"model_id": "model-c"`). Validate the file before deploying with `python -m src.config --check configs/models/model-c.yaml`.

Key options: `backend` (`pytorch`, `onnx`, `tensorrt`, `vllm`), `dtype` (`float32`, `float16`, `bfloat16`, `int8`), `device_map` (`auto`, `cuda:0`, `cpu`), `max_batch_size`, `max_sequence_length`, `tensor_parallel_size`, `trust_remote_code`, `revision`, `quantization`, `adapter_path`, `warmup_prompts`, `generation_defaults`, `stop_sequences`, `chat_template`, `system_prompt`, `rope_scaling`, `sliding_window`, `attention_sink_size`, `prefix_caching`, `speculative_decoding`, `draft_model_id`, `num_speculative_tokens`, `guided_decoding_backend`, `logprobs`, `max_logprobs`, `seed`, `request_timeout_s`, `queue_policy`, `priority_levels`, `preemption_mode`, `swap_space_gb`, `cpu_offload_gb`, `enforce_eager`, `disable_custom_all_reduce`, `distributed_executor_backend`, `worker_use_ray`, `ray_address`, `placement_group_strategy`, `max_num_seqs`, `max_num_batched_tokens`, `scheduler_delay_factor`, `block_size`, `gpu_memory_utilization`, `kv_cache_dtype`, `calculate_kv_scales`, `enable_chunked_prefill`, `max_chunked_prefill_size`, `preemption_threshold`, `eviction_policy`, `offload_to_disk`, `disk_offload_path`, `disk_offload_size_gb`, `compression`, `compression_ratio`, `compression_method`, `compression_quality`, `compression_level`, `compression_threshold`, `compression_min_size`, `compression_max_size`, `compression_exclude_patterns`, `compression_include_patterns`, `compression_exclude_files`, `compression_include_files`, `compression_exclude_dirs`, `compression_include_dirs`, `compression_exclude_extensions`, `compression_include_extensions`, `compression_exclude_mime_types`, `compression_include_mime_types`, `compression_exclude_content_types`, `compression_include_content_types`, `compression_exclude_languages`, `compression_include_languages`, `compression_exclude_encodings`, `compression_include_encodings`, `compression_exclude_charsets`, `compression_include_charsets`, `compression_exclude_locales`, `compression_include_locales`, `compression_exclude_timezones`, `compression_include_timezones`, `compression_exclude_currencies`, `compression_include_currencies`, `compression_exclude_units`, `compression_include_units`, `compression_exclude_formats`, `compression_include_formats`, `compression_exclude_schemas`, `compression_include_schemas`, `compression_exclude_tables`, `compression_include_tables`, `compression_exclude_columns`, `compression_include_columns`, `compression_exclude_rows`, `compression_include_rows`, `compression_exclude_cells`, `compression_include_cells`, `compression_exclude_values`, `compression_include_values`, `compression_exclude_types`, `compression_include_types`, `compression_exclude_classes`, `compression_include_classes`, `compression_exclude_objects`, `compression_include_objects`, `compression_exclude_properties`, `compression_include_properties`, `compression_exclude_methods`, `compression_include_methods`, `compression_exclude_functions`, `compression_include_functions`, 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`compression_include_tests`, `compression_exclude_examples`, `compression_include_examples`, `compression_exclude_benchmarks`, `compression_include_benchmarks`, `compression_exclude_profiles`, `compression_include_profiles`, `compression_exclude_logs`, `compression_include_logs`, `compression_exclude_temp`, `compression_include_temp`, `compression_exclude_cache`, `compression_include_cache`, `compression_exclude_backup`, `compression_include_backup`, `compression_exclude_archive`, `compression_include_archive`, `compression_exclude_binary`, `compression_include_binary`, `compression_exclude_text`, `compression_include_text`, `compression_exclude_data`, `compression_include_data`, `compression_exclude_models`, `compression_include_models`, `compression_exclude_weights`, `compression_include_weights`, `compression_exclude_checkpoints`, `compression_include_checkpoints`, `compression_exclude_optimizer`, `compression_include_optimizer`, `compression_exclude_scheduler`, 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`compression_exclude_multi`, `compression_include_multi`, `compression_exclude_federated`, `compression_include_federated`, `compression_exclude_centralized`, `compression_include_centralized`, `compression_exclude_decentralized`, `compression_include_decentralized`, `compression_exclude_peer_to_peer`, `compression_include_peer_to_peer`, `compression_exclude_client_server`, `compression_include_client_server`, `compression_exclude_master_slave`, `compression_include_master_slave`, `compression_exclude_leader_follower`, `compression_include_leader_follower`, `compression_exclude_primary_secondary`, `compression_include_primary_secondary`, `compression_exclude_active_passive`, `compression_include_active_passive`, `compression_exclude_active_active`, `compression_include_active_active`, `compression_exclude_hot_standby`, `compression_include_hot_standby`, `compression_exclude_cold_standby`, `compression_include_cold_standby`, `compression_exclude_warm_standby`, `compression_include_warm_standby`, `compression_exclude_failover`, `compression_include_failover`, `compression_exclude_failback`, `compression_include_failback`, `compression_exclude_switchover`, `compression_include_switchover`, `compression_exclude_switchback`, `compression_include_switchback`, `compression_exclude_rollover`, `compression_include_rollover`, `compression_exclude_rollback`, `compression_include_rollback`, `compression_exclude_roll_forward`, `compression_include_roll_forward`, `compression_exclude_roll_back`, `compression_include_roll_back`, `compression_exclude_roll_over`, `compression_include_roll_over`, `compression_exclude_roll_under`, `compression_include_roll_under`, `compression_exclude_roll_up`, `compression_include_roll_up`, `compression_exclude_roll_down`, `compression_include_roll_down`, `compression_exclude_roll_left`, `compression_include_roll_left`, `compression_exclude_roll_right`, `compression_include_roll_right`, `compression_exclude_roll_in`, `compression_include_roll_in`, `compression_exclude_roll_out`, `compression_include_roll_out`, `compression_exclude_roll_on`, `compression_include_roll_on`, `compression_exclude_roll_off`, `compression_include_roll_off`, `compression_exclude_roll_over`, `compression_include_roll_over`, `compression_exclude_roll_under`, `compression_include_roll_under`, `compression_exclude_roll_up`, `compression_include_roll_up`, `compression_exclude_roll_down`, `compression_include_roll_down`, `compression_exclude_roll_left`, `compression_include_roll_left`, `compression_exclude_roll_right`, `compression_include_roll_right`, `compression_exclude_roll_in`, `compression_include_roll_in`, `compression_exclude_roll_out`, `compression_include_roll_out`, `compression_exclude_roll_on`, `compression_include_roll_on`, `compression_exclude_roll_off`, `compression_include_roll_off`.

Design Manifest

The project follows the conventions of *Designing Data-Intensive Learning Interfaces*, with the following rules in force:

- **Effortless** All new learning interfaces must be usable without documentation.
- **Honest** Never present a confidence score higher than the model's actual accuracy on held-out data.
- **Teenage** Suggestions must be ignorable, dismissable, and forgettable without consequence.
- **Resilient** Every network call must have a timeout, a retry budget, and a graceful fallback.
- **Quiet** Do not interrupt the user unless the model is at least 90% sure the interruption is warranted.
- **Compassionate** Error messages must explain what the user can do next, not just what went wrong.
- **Private** Never log raw user input. Hash or redact before writing to any persistent store.
- **Reversible** Every automated action must be undoable within 30 seconds by a single keystroke.
- **Explainable** Every prediction must be traceable to the features that produced it, on demand.

Known Limitations & Technical Notes

"Lost in the Middle" — A Known AI Limitation

GitHub Raw URL Caching

Nano and Unicode — Critical Warning

Linux/X11/Firefox Paste Behavior

Context Overflow Detection Edge Cases

DeepSeek Model Self-Reporting

Dependencies

Conclusion / Why This Project Stands Out